INTRODUCTION TO DATABASE MANAGEMENT SYSTEMS

Module 1 – CPE011 DBMS

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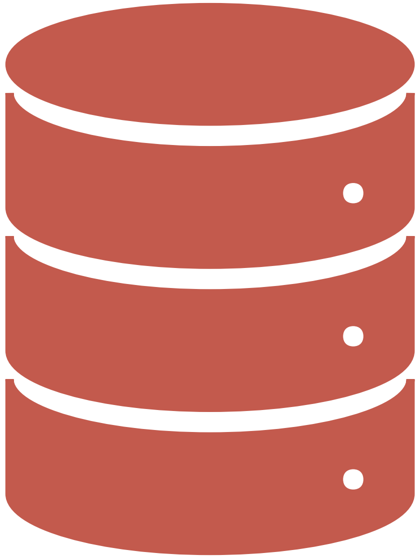
OBJECTIVES:



At the end of this module, students would be able to:

1. Explain the importance of databases, the different types of databases, and give its main functions and advantages
2. Compare Manual and Computerized File Systems to design database systems.
3. Analyze how the systems and database development life cycle can be used in designing database systems.
4. Demonstrate how to create , modify and delete database
5. Demonstrate how to insert, update and delete records to the database

WHAT IS DATABASE?

A database is an ***organized collection of structured information***, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS). Together, the data and the DBMS, along with the applications that are associated with them, are referred to as a database system, often shortened to just database.

Data within the most common types of databases in operation today is typically modeled in rows and columns in a series of tables to make processing and data querying efficient. The data can then be easily accessed, managed, modified, updated, controlled, and organized. Most databases use structured query language (SQL) for writing and querying data.

WHAT IS STRUCTURED QUERY LANGUAGE (SQL)?

SQL is a programming language used by nearly all relational databases *to query, manipulate, and define data, and to provide access control*.

SQL was first developed at IBM in the 1970s with Oracle as a major contributor, which led to implementation of the SQL ANSI standard, SQL has spurred many extensions from companies such as IBM, Oracle, and Microsoft. Although SQL is still widely used today, new programming languages are beginning to appear.

DATABASE SYSTEM APPLICATIONS

Enterprise Information

- Sales
- Accounting
- Human Resources
- Manufacturing
- Online Retailers

DATABASE SYSTEM APPLICATIONS

Banking and Finance

- Banking
- Credit Card Transactions
- Finance

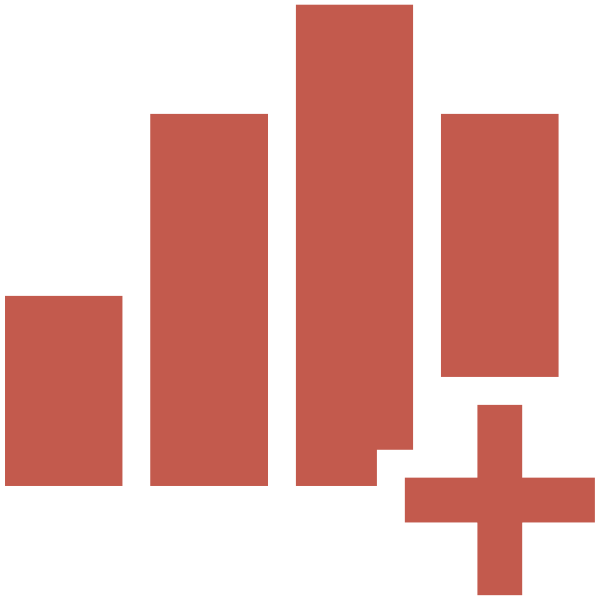
DATABASE SYSTEM APPLICATIONS

Other Fields

- Universities
- Airlines
- Telecommunications

DATA VS INFORMATION

- **Data** consists of the raw facts.
- **Information** is the result of processing raw data to reveal its meaning.



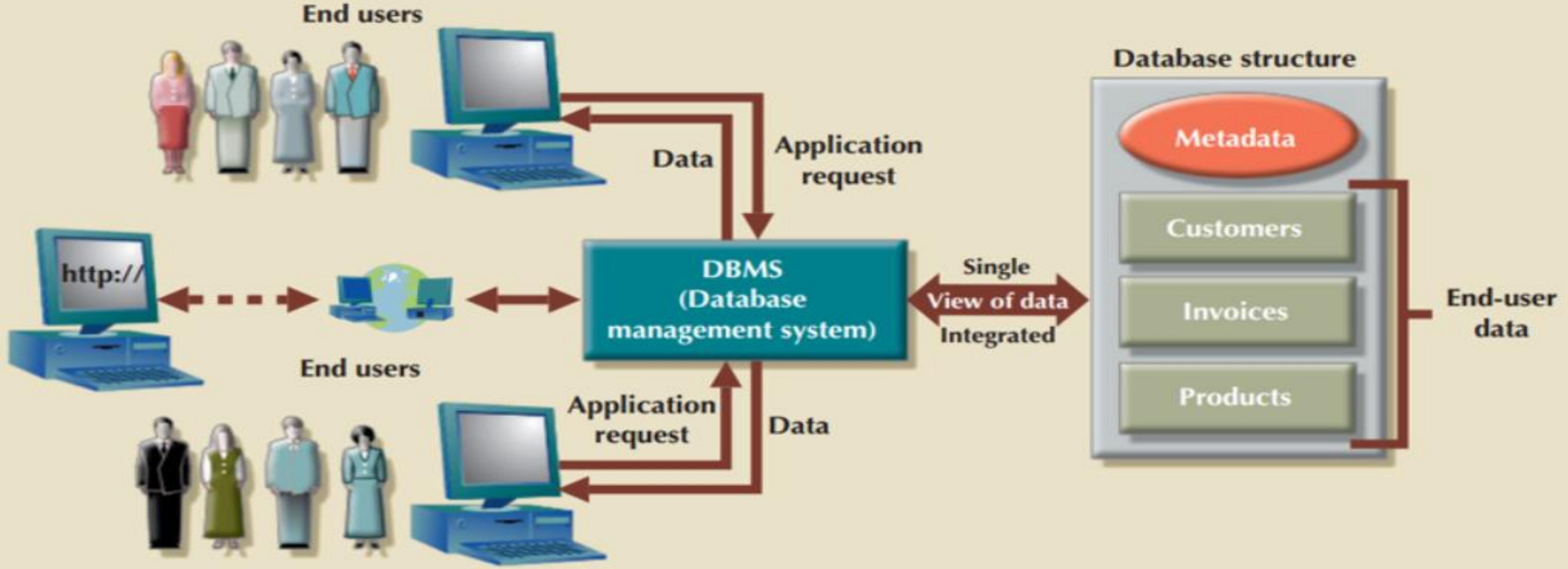
FUNCTIONS AND ADVANTAGES OF DBMS

A **database** is a shared, integrated computer structure that stores a collection of the following:

- **End-user data**—that is, raw facts of interest to the end user.
- **Metadata**, or data about data, through which the end-user data is integrated and managed.

A **database management system (DBMS)** is a collection of programs that manages the database structure and controls access to the data stored in the database.

FUNCTIONS AND ADVANTAGES OF DBMS



WHAT IS A DATABASE MANAGEMENT SYSTEM (DBMS)?

What is a database management system (DBMS)?

A database typically requires a **comprehensive database software program** known as a database management system (DBMS). A DBMS serves as an interface between the database and its end users or programs, allowing users to retrieve, update, and manage how the information is organized and optimized. A DBMS also facilitates oversight and control of databases, enabling a variety of administrative operations such as performance monitoring, tuning, and backup and recovery.

Some examples of popular database software or DBMSs include MySQL, Microsoft Access, Microsoft SQL Server, FileMaker Pro, Oracle Database, and dBASE.

ADVANTAGES OF DATABASE MANAGEMENT SYSTEM (DBMS)

- Improved data sharing
- Improved data security
- Better data integration
- Minimized data inconsistency
- Improved decision making
- Increased end-user productivity

FUNCTIONS OF DATABASE MANAGEMENT SYSTEM (DBMS)

- Data dictionary management
- Data storage management
- Data transformation and presentation
- Security management

FUNCTIONS OF DATABASE MANAGEMENT SYSTEM (DBMS)

- Multiuser access control
- Backup and recovery management
- Data integrity management
- Database access languages and application programming interfaces
- Database communication interfaces

TYPES OF DATABASES

ACCORDING TO THE NUMBER OF USERS

- Single user database
- Multiuser database

TYPES OF DATABASES

BASED ON LOCATION

- Centralized database
- Distributed database

TYPES OF DATABASES

ACCORDING TO THE TYPE OF DATA

- General purpose database
- Discipline-specific database

TYPES OF DATABASES

BASED ON INTENDED DATA USAGE

- Operational or Transactional database
- Analytical database

TYPES OF DATABASES

ACCORDING TO THE DEGREE WITH WHICH DATA IS STRUCTURED

- Unstructured database
- Structured database
- Semi-structured database

TYPES OF DATABASES

RISE OF BIG DATA AND NoSQL

- NoSQL Database

WHAT’S THE DIFFERENCE BETWEEN A DATABASE AND A SPREADSHEET?

Databases and spreadsheets (such as Microsoft Excel) are both convenient ways to store information. The primary differences between the two are:

- How the data is stored and manipulated
- Who can access the data
- How much data can be stored

WHAT'S THE DIFFERENCE BETWEEN A DATABASE AND A SPREADSHEET?

Spreadsheets

- originally designed for one user
- great for a single user or small number of users who don't need to do a lot of complicated data manipulation.

Databases

- designed to hold much larger collections of organized information
- allow multiple users at the same time to quickly and securely access and query the data using highly complex logic and language

IMPORTANCE OF DATABASE DESIGN

Database design refers to the activities that focus on the design of the database structure that will be used to store and manage end-user data.

EXAMPLE OF POORLY DESIGNED DATABASE

Why are there blanks in rows 9 and 10?

How to produce an alphabetical listing of employees?

How to count how many employees are certified in Basic Database Manipulation?

Is Basic Database Manipulation the same as Basic DB Manipulation?

What if an employee acquires a fourth certification?

Do we add another column?

ID	ENum	Name	Title	HireDate	Skill1	Skill1Date	Skill2	Skill2Date	Skill3	Skill3Date
1	02345	Brian Oates	DBA	2/14/1995	Basic Database Management	2/14/2002	Advanced Database Management	2/14/2005	Basic Web Design	8/9/2003
2	08273	Marco Bienz	Analyst	7/28/2006	Basic Web Design	3/8/2009	Advance Process Modeling	8/19/2012		
3	06234	Jasmine Patel	Programmer	8/10/2005	Basic Web Design	8/10/2007	Advanced C# programming	8/10/2007	Basic DB manipulation	1/29/2012
4	03373	Franklin Johnson, Jr.	Purchasing Agent	3/15/2002	Advanced Spreadsheets	6/20/2011				
5	13567	Almond, Robert	Analyst	9/30/2012	Basic Process Modeling	9/30/2014	Basic Database Design	5/23/2015		
6	10282	Richardson, Amanda	Clerk	4/11/2011						
7	09382	Susan Mathis	Database Programmer	8/2/2010	Basic DB Design	8/2/2012	Basic Database Manipulation	8/2/2012	Advanced DB Manipulation	5/1/2013
8	14311	Duong, Lee	Programmer	9/1/2014	Basic Web Design	9/1/2016				
9					Master Database Programming					
10					Basic Spreadsheets					
11	09002	Wade Gaither	Clerk	5/20/2010	Advanced Spreadsheets	5/16/2013	Basic Web Design	5/16/2013		
12	13383	Raymond F. Matthews	Programmer	3/12/2012	Basic C# Programming	3/12/2014				
13	09283	Chavez, Juan	Clerk	7/4/2010						
14	04893	Patricia Richards	DBA	6/11/2004	Advanced Database Management	6/11/2006	Advanced Database Manipulation	9/20/2012		
15	13932	Lee, Megan	Programmer	9/29/2013						

PROBLEMS WITH FILE SYSTEM DATA PROCESSING

- Lengthy development times
- Difficulty of getting quick answers
- Complex system administration
- Lack of security and limited data sharing
- Extensive programming

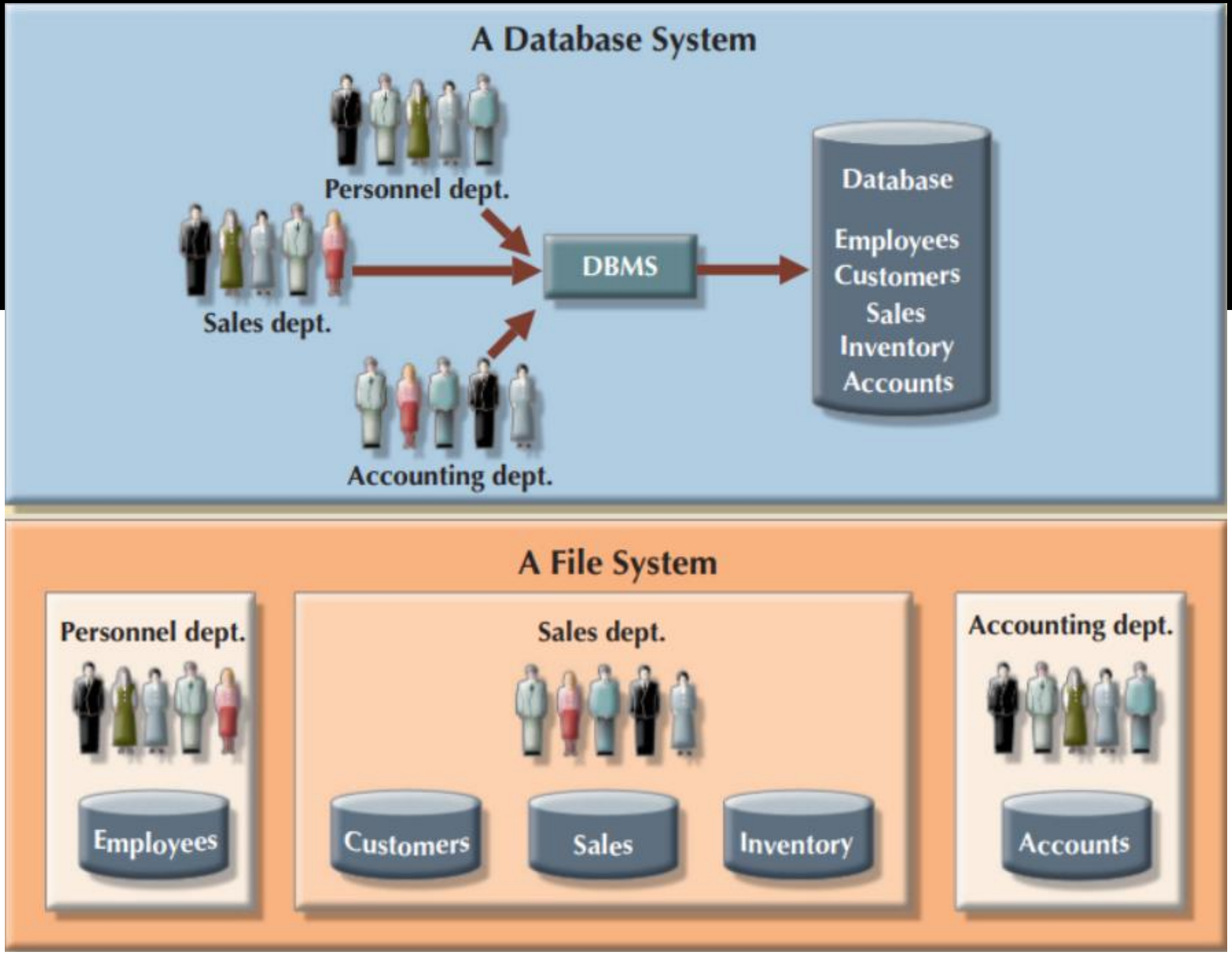
OTHER PROBLEMS

- Structural and Data Dependence
- Data Redundancy
  - Poor data security
  - Data inconsistency
  - Data-entry errors
  - Data integrity errors

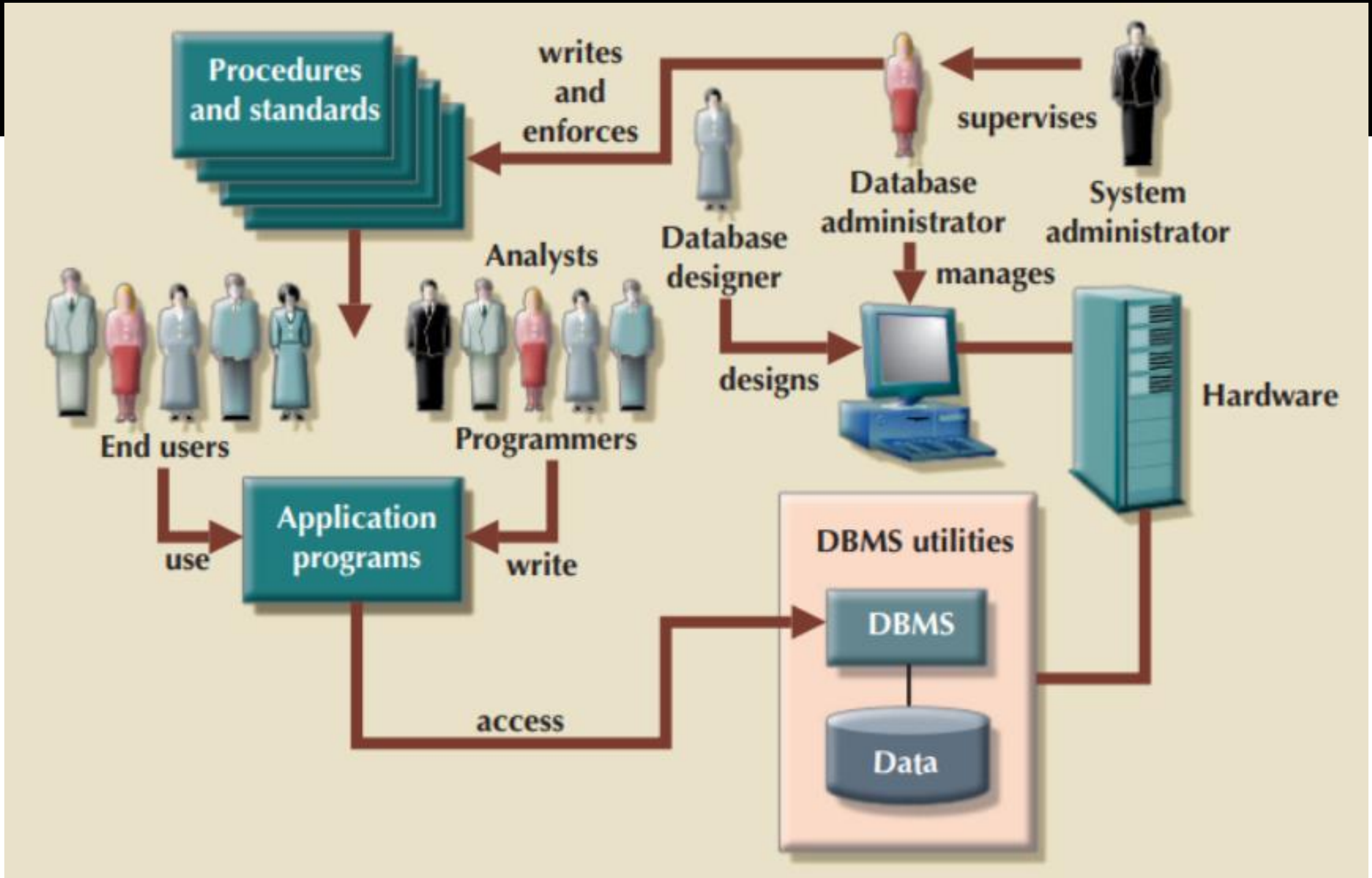
OTHER PROBLEMS

- Data Anomalies
  - Update anomalies
  - Insertion anomalies
  - Deletion anomalies

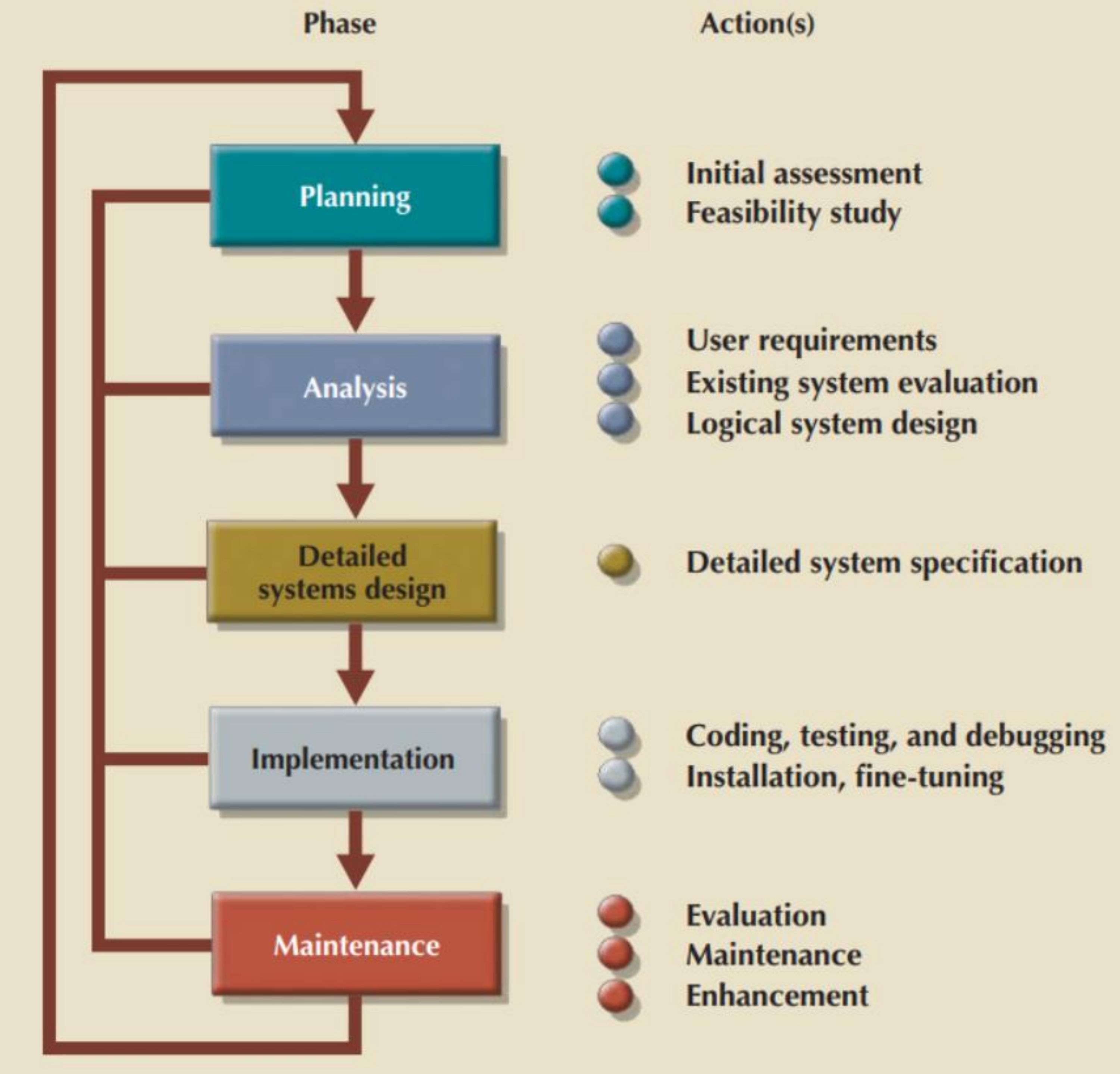
CONTRASTING DATABASE AND FILE SYSTEMS



DATABASE SYSTEM ENVIRONMENT



SDLC


Phase
Action(s)
Planning
Initial assessment
Feasibility study
Analysis
User requirements
Existing system evaluation
Logical system design
Detailed systems design
Detailed system specification
Implementation
Coding, testing, and debugging
Installation, fine-tuning
Maintenance
Evaluation
Maintenance
Enhancement

he company situation
oblems and constraints
jectives
ope and boundaries

e conceptual design
tware selection
e logical design
e physical design

DBMS
e database(s)
onvert the data

atabase
the database
he database and its application program

he required information flow

changes

DATABASE CAREER OPPORTUNITIES

JOB TITLE	DESCRIPTION	SAMPLE SKILLS REQUIRED
Database Developer	Create and maintain database-based applications	Programming, database fundamentals, SQL
Database Designer	Design and maintain databases	Systems design, database design, SQL
Database Administrator	Manage and maintain DBMS and databases	Database fundamentals, SQL, vendor courses
Database Analyst	Develop databases for decision support reporting	SQL, query optimization, data warehouses
Database Architect	Design and implementation of database environments (conceptual, logical, and physical)	DBMS fundamentals, data modeling, SQL, hardware knowledge, etc.
Database Consultant	Help companies leverage database technologies to improve business processes and achieve specific goals	Database fundamentals, data modeling, database design, SQL, DBMS, hardware, vendor-specific technologies, etc.
Database Security Officer	Implement security policies for data administration	DBMS fundamentals, database administration, SQL, data security technologies, etc.
Cloud Computing Data Architect	Design and implement the infrastructure for next-generation cloud database systems	Internet technologies, cloud storage technologies, data security, performance tuning, large databases, etc.

INTRODUCTION TO SQL

CATEGORIES:

- Data Manipulation Language
- Data Definition Language
- Transaction Control Language
- Data Control Language